

STA 4700 - Categorical Data Analysis

Dr. Mun

Final Project: Customer Churn Factors

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Data Source:

<https://www.kaggle.com/datasets/adrianvinueza/gym-customers-features-and-churn>

Abstract:

Customer churn—the loss of clients or subscribers—poses a significant threat to business sustainability and growth. This study demonstrates how combining statistical inference with machine learning techniques can enhance churn prediction and guide retention-focused decision-making. Homogeneity tests with residual analysis reveal categorical variables that significantly differ between Churn groups. A logistic regression model with Lasso regularization is applied to predict churn and isolate the most influential predictors, achieving a balance between accuracy and interpretability. The approach provides actionable insights for identifying at-risk customers and supports data-driven strategies to reduce churn.

Introduction to Dataset, Data Manipulation, and Initial Findings:

The dataset, sourced from Kaggle, contains 4,000 observations and 14 variables, including five numeric features, eight categorical features, and one response variable—Churn. The variables include: Gender, Near Location, Partner, Promo Friends, Phone, Contract_Period, Group_Visits, Age, Average Additional Charges, Months to End of Contract, Lifetime, Average Class Frequency, Current Class Frequency, and Churn. There were no missing values in the dataset.

Several preprocessing steps were taken to enhance the interpretability and modeling of the data. The Contract_Period variable was recoded to better reflect customer status. Average Additional Charges were rounded to the nearest cent. The Months to End of Contract variable was binned into three categories: Renew (1 month), Upcoming (2–5 months), and Later (6–12 months), in order to reduce variance and highlight renewal intention. Similarly, Average Class Frequency was binned to reflect usage tiers: Beginner (0–1.5 classes), Active (1.5–3 classes), and Regular (more than 3 classes), acknowledging that fractional class attendance is less meaningful. A new variable, Class_Deviance, was created by subtracting Current Class Frequency from Average Class Frequency, aiming to capture deviations in recent attendance behavior and potential churn risk. Current Class Frequency was excluded from the analysis.

Summary statistics show that the dataset is balanced in terms of gender (49% Group X, 51% Group Y). The majority of users have a phone number on file (90%), live near the gym (85%), and are employed at a partnered company (49%). Promo friend sign-ups are common (69%), although only 41% have brought a friend. Regarding Contract_Period, 55% of customers are on a monthly plan, 24% on an annual plan, and 21% on a semi-annual plan. For the Contract Tier, 55% are due for renewal within one month, 39% within two to five months, and 6% in six or more months. The Class History

breakdown shows 37% are beginners, 50% are active users, and 13% are regulars. No Strong correlations occur between the numeric variables ruling out multicollinearity.

Preliminary Analysis:

Since the marginal totals are fixed, a test of homogeneity can be applied to determine whether the distribution of categorical variables differs across Churn groups. This approach helps identify which categorical features are associated with customer churn and meaningfully contribute to distinguishing between groups.

a. Comparing the Response Variable with Categorical Variables

Table of Categorical Variables and P-values of G-Test with Churn Groups

Column <chr>	P_Value <dbl>
Gender	9.642976e-01
Location	4.773959e-15
Partner_Company	0.000000e+00
Friends_Promo	0.000000e+00
Contract_Type	0.000000e+00
Bring_Friends	0.000000e+00
Contract_Renewal	0.000000e+00
Class_History	0.000000e+00

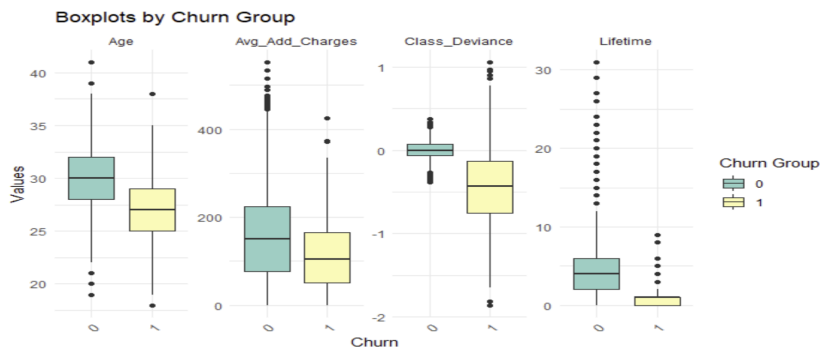
The G-test is used to evaluate whether the distribution of each categorical variable differs significantly across Churn groups. A key assumption of the expected count in each cell exceeds five is met. The null hypothesis states that the distributions of the explanatory variables are homogeneous across Churn groups, while the alternative hypothesis suggests that at least one group's distribution deviates from the others.

Based on the table of p-values from the G-test, which compares each explanatory variable to Churn status, we can identify which variables are statistically significant. A significant result indicates a violation of homogeneity and suggests meaningful differences in how Churn groups respond, providing insight into the factors associated with customer cancellations.

For instance, the variable Gender has a p-value of 0.96, indicating no statistically significant difference in its distribution across Churn groups. In contrast, all other variables have p-values below 0.05—many below 0.001—providing strong evidence against the null hypothesis. These results imply that the distributions of the Churn groups differ significantly across the categorical variables and may be predictive of customer retention or attrition.

To further understand these differences, residual analysis will be performed to identify which response categories contribute most to the observed deviations. This will help pinpoint areas of strength among retained customers and highlight opportunities for improving retention among those at risk of churning.

b. Comparing the Response Variable with Numerical Variables



Plotting the Churn groups across the numerical variables reveals key differences in their distributions. For Age, there is a slight distinction between groups: customers who stayed tend to have a higher median age and a broader spread, whereas those who churned are more concentrated among younger individuals, particularly under 30.

The variable Average Additional Charges shows a more pronounced separation. Customers who churn tend to incur lower additional charges, while those who stay often have significantly higher charges. This suggests that engagement with extra services or products may be associated with retention. A deeper exploration into what these charges represent—such as specific classes, amenities, or merchandise—could provide actionable insights into consumer behavior and churn drivers.

For Class Deviance, a clear pattern emerges: customers who churn tend to exhibit negative deviance, meaning they attend fewer classes than their average, which may indicate a loss of interest or declining motivation. Interestingly, there is also evidence that customers who suddenly attend more classes than their average may also be at risk of churning—potentially suggesting a short-term trial or last effort before discontinuing their membership. In contrast, customers who stay typically show stable or increasing attendance over time. It is recommended to implement targeted outreach, such as motivational reminders or personalized messages, both when attendance drops below a customer's average and when an unexpected spike in participation occurs, to help reinforce engagement and prevent churn.

Lifetime also reveals important distinctions. Most retained customers have been with the gym for four to six months, while those who churn often do so within the first few months and rarely stay longer than ten. This highlights a vulnerable period early in the customer lifecycle. Exploring the reasons behind early churn—such as unmet expectations or lack of engagement—could guide new onboarding policies or incentive programs aimed at encouraging longer membership durations.

In summary, while most numerical variables show meaningful differences between Churn groups, Age appears to have a weaker and less direct relationship. In contrast, Average Additional Charges, Class Deviance, and Lifetime demonstrate stronger associations with customer retention and should be prioritized in subsequent analyses.

Main Analysis:

In the main analysis, we will begin with residual analysis to interpret how the categorical variables influence the distribution of Churn groups. Following this, we will develop a logistic regression model to predict future churn.

a. Residual Analysis

This analysis identifies where the distributions are skewed by interpreting standardized residuals greater than ± 3.0 as strong indicators of over- or underrepresentation within the Churn groups.

The residuals for Gender reveal the distribution between the Churn groups is not significantly different, with all residuals falling within ± 0.5 —suggesting no notable imbalance. In contrast, Location shows a strong overrepresentation of customers who live nearby and stayed (residual = 8.1), suggesting that proximity to the gym may be an indicator of retention. Customers affiliated with a Partner Company are also significantly overrepresented in the “stay” group (residual = 10), suggesting that organizational partnerships may contribute to long-term membership. The Friends Promo similarly appears effective, as customers who enrolled in this promotion were highly overrepresented among those who stayed (residual = 10.2). Regarding Contract Type, strong patterns emerge: customers on Annual and Semi-Annual contracts are overrepresented among those who stayed (residuals = 19.4 and 10.3, respectively), while those on Monthly contracts are heavily underrepresented (residual = -25.1). This underscores a clear link between longer contract durations and improved retention. Customers who Bring Friends also exhibit an overrepresentation in the “stay” group (residual = 11), indicating that social engagement may play a role in continued membership. Initiatives such as buddy passes or group fitness classes may help reinforce this behavior and enhance retention. Finally, the Class History variable highlights that Active and Regular participants are more likely to stay (residuals = 1.8 and 13.2), while Beginners are underrepresented (residual = -11.1). This suggests that newer or less engaged members may be at greater risk of churn, possibly due to limited familiarity with gym offerings or lower initial motivation.

b. Logistic Regression Model

To predict customer churn and identify influential attributes, we will build a logistic regression model. The dataset will be split into training (70%) and testing (30%) subsets. The training set will be used to develop the model, while the test set will assess its predictive performance.

Based on the previous analysis, all variables—except Gender—were found to contribute meaningfully to Churn classification and will be included in the modeling process. To enhance model generalizability and reduce overfitting, we will apply Lasso and Ridge regularization techniques. These methods introduce penalties to the regression coefficients, in order to minimize the log-likelihood, allowing for variable selection (Lasso) and parameter shrinkage (Ridge). We will use 5-fold cross-validation

during model training to optimize regularization parameters and ensure robust model performance.

Model evaluation will focus on key classification metrics, including accuracy, area under the ROC curve (AUC), sensitivity, and specificity. Lastly, we will interpret the resulting coefficients to understand the influence of each predictor on the likelihood of customer churn.

I. Model Selection

Name_of_Model<chr>	Accuracy<dbl>	AUC<dbl>	Sensitivity<dbl>	Specificity<dbl>
Lasso	0.9374	0.9753	0.9841	0.8082
Ridge	0.9291	0.9735	0.9807	0.7862
Ridge No Gender	0.9308	0.9737	0.9807	0.7925

3 rows

After developing three models—Lasso using all variables, Ridge using all variables, and Ridge excluding Gender—we assessed model performance using key metrics. Lasso achieved the highest accuracy on the test set at 93.74%, with both Ridge models trailing by less than 1%. For AUC, all models performed similarly, with differences within 0.0018, and Lasso again showed the best result. Lasso also achieved the highest sensitivity at 0.9841, with both Ridge models within 0.0035 of that value. Lastly, Lasso outperformed in specificity, exceeding 80%.

Based on these metrics, Lasso emerges as the best overall model. Its superior accuracy on the test set suggests it is well-generalized and likely to perform reliably on new data. The higher AUC supports its strong discriminative ability, especially in identifying customers at risk of churn. Its high sensitivity is particularly important, as it ensures we effectively identify those likely to leave—minimizing missed opportunities for targeted intervention. Furthermore, the strong specificity indicates that the model can accurately recognize customers who are likely to stay, reducing the risk of over-targeting loyal members with unnecessary promotions or outreach.

II. Evaluating Parameters

Through analyzing the parameter values we can see which variables have the greatest influences on predicting the Churn groups.

Parameter<chr>	Lasso<dbl>	Parameter<chr>	Lasso<dbl>
(Intercept)	5.8974	Contract_RenewalRenew	0.0000
Gender1	0.0000	Contract_RenewalUpcomming	0.0000
Location1	0.0000	Lifetime	-0.5425
Partner_Company1	0.0000	Class_HistoryBeginner	0.2843
Friends_Promo1	-0.1525	Class_HistoryRegular	-0.6958
Contract_TypeMonthly	1.3905	Class_Deviance	-3.5025
Contract_TypeSemi-Annual	0.0000		
Bring_Friends1	-0.3416		
Age	-0.2308		
Avg_Add_Charges	-0.0023		

Based on the final Lasso model, several variables were excluded due to their lack of predictive value. These include Gender, Location, Partner Company, Contract Type – Semi-Annual, and Contract Renewal. In this context, a model that predicts churn

with a smaller subset of variables offers clearer interpretation of each variable's direct impact on churn behavior. As more data becomes available and the importance of variables shifts over time, the Ridge model can be reconsidered to capture broader, more nuanced influences and uncover previously unobserved relationships.

The exclusion of these variables indicates they were not statistically significant in predicting churn in the current dataset. Consistent with earlier findings, we observe that the longer a customer remains subscribed, the lower their likelihood of churning. Additionally, customers in the Beginner class tier show higher churn risk, whereas those who advance to the Regular tier are less likely to leave—reaffirming the importance of sustained class engagement. The variable Class Deviance also aligns with prior analysis: customers who churn typically show higher negative deviations from their average class attendance.

With the model now built, fine-tuned, and interpreted, it can be used to predict future churn cases. Furthermore, we can adjust the decision threshold to be more sensitive, enabling the interpretation of predicted probabilities as a loyalty score. This approach allows us to proactively identify customers who may be on the verge of churning, offering an opportunity to intervene before they disengage entirely.

Results and Recommendations:

Findings from this analysis emphasize the importance of engagement and contract commitment, while demographic variables show limited influence. The model supports targeted outreach by identifying customers at risk of churning, enabling data-driven interventions to policy and marketing promotions. The results from the test of homogeneity and the Lasso model provide two complementary but important perspectives. The homogeneity test, supported by residual analysis, highlights where the Churn groups differ or align across categorical variables, offering insight into behavioral patterns and imbalances. In contrast, the Lasso model identifies which variables are most predictive of customer churn, helping isolate the most influential features for targeted intervention.

Together, these analyses enhance our understanding of churn behavior and support data-driven policy decisions aimed at reducing churn. To further refine this understanding, additional exploratory data analysis is recommended to investigate potential confounding factors and relationships within subgroups—for example, how Age interacts with Additional Charges. Incorporating more detailed consumer data—such as the number of services used or products purchased—could also improve the accuracy and depth of churn modeling.

Finally, the predictive capabilities of the model can be leveraged to identify customers at risk of churning, allowing for strategic outreach through personalized promotions or supportive messaging. These proactive efforts can influence customer decisions and strengthen retention.